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$$f(x) = \frac{e^{-x}}{(1+e^{-x})^2}, \quad -\infty < x < \infty$$

(1)

$$\int_{-\infty}^{\infty} \frac{e^{-x}}{(1+e^{-x})^2} dx$$

$$1+e^{-x} = t$$

x	$-\infty$	$\rightarrow$	∞
t	∞	$\rightarrow$	1

$$\frac{dt}{dx} = -e^{-x}$$

$$-e^{-x} \cdot dx = dt$$

$$\int_{\infty}^1 \frac{-1}{t^2} dt$$

$$= \int_1^{\infty} \frac{1}{t^2} dt$$

$$= \left[-t^{-1} \right]_1^{\infty} = -\frac{1}{\infty} + 1 = 1$$

(2)

$$F(x) = \int_{-\infty}^x \frac{e^{-x'}}{(1+e^{-x'})^2} dx'$$

$$1+e^{-x'} = t$$

x'	$-\infty$	$\rightarrow$	x
t	∞	$\rightarrow$	$1+e^{-x}$

$$F(x) = \int_{\infty}^{1+e^{-x}} \frac{-1}{t^2} dt$$

$$= \int_{1+e^{-x}}^{\infty} \frac{1}{t^2} dt$$

$$= \left[-t^{-1} \right]_{1+e^{-x}}^{\infty}$$

$$= \frac{1}{1+e^{-x}}$$

$$f(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{\frac{1}{e^x}}{\left(1+\frac{1}{e^x}\right)^2} = \frac{\frac{1}{e^x}}{\left(\frac{e^x+1}{e^x}\right)^2}$$

$$= \frac{e^x}{(1+e^x)^2} = f(-x)$$

$$\frac{\frac{1}{e^x}}{\frac{e^x+1}{e^x}} = \frac{1}{e^x+1}$$

(3)

$$f_u(u) = f_x\left(\ln \frac{1-u}{u}\right) \cdot \left| \frac{d}{du} \ln \frac{1-u}{u} \right|$$

$$u = g(x) = \frac{e^{-x}}{1+e^{-x}}$$

$$\frac{1}{u} = \frac{1+e^{-x}}{e^{-x}}$$

$$= 1+e^{-x}$$

$$e^{-x} = \frac{1}{u} - 1$$

$$x = \ln \frac{1-u}{u}$$

$$g'(u) = \ln \frac{1-u}{u}$$

$$\left(\frac{t}{1+t}\right)' = \frac{(1+t)-t}{(1+t)^2} = \frac{1}{(1+t)^2}$$

$$f_u(u) = \frac{e^{-\ln \frac{1-u}{u}}}{\left(1+e^{-\ln \frac{1-u}{u}}\right)^2} \times \left| \frac{u}{1-u} \cdot \frac{-u-(1-u)}{u^2} \right|$$

$$= \frac{\frac{u}{1-u}}{\left(1+\frac{u}{1-u}\right)^2} \times \left| \frac{-1}{u(1-u)} \right|$$

$$= \frac{\frac{u}{1-u}}{\left(\frac{1}{1-u}\right)^2} \times \frac{1}{u(1-u)}$$

$$= \frac{1}{1}$$

(4)

$$Y = |X| \rightarrow Y = \begin{cases} X & x > 0 \\ -X & 0 > x \end{cases}$$

 $x < 0 \text{ or } x \geq$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(-X \leq y) \\ &= P(X \geq -y) \\ &= \int_{-y}^0 f_X(x) dx \\ &= \int_{-y}^0 \frac{e^{-x}}{(1+e^{-x})^2} dx \\ &= \left[\frac{1}{1+e^{-x}} \right]_{-y}^0 \\ &= \frac{1}{1+1} - \frac{1}{1+e^y} \\ &= \frac{1}{2} - \frac{1}{1+e^y} \\ f_Y(y) &= \frac{d}{dy} \left(\frac{1}{2} - \frac{1}{1+e^y} \right) \\ &= + \frac{e^y}{(1+e^y)^2} \end{aligned}$$

 $x \geq 0$

$$\begin{aligned} F_Y(y) &= P(X \leq y) \\ &= \int \end{aligned}$$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(|X| \leq y) \\ &= P(-y \leq X \leq y) \\ &= \int_{-y}^y f_X(x) dx \\ &= \int_{-y}^y \frac{e^{-x}}{(1+e^{-x})^2} dx \\ &= \left[\frac{1}{1+e^{-x}} \right]_{-y}^y \\ &= \frac{1}{1+e^{-y}} - \frac{1}{1+e^y} = \frac{e^y - 1}{1+e^y} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} \left(\frac{1}{1+e^{-y}} - \frac{1}{1+e^y} \right) \\ &= \frac{e^{-y}}{(1+e^{-y})^2} + \frac{e^y}{(1+e^y)^2} \\ &= \frac{\frac{1}{e^y}}{\left(\frac{e^y+1}{e^y} \right)^2} = \frac{1}{e^y} \times \frac{(e^y)^2}{(e^y+1)^2} \\ &= \frac{e^y}{(e^y+1)^2} \\ &= \frac{2e^y}{(1+e^y)^2} \end{aligned}$$

$$\begin{aligned} \lambda(y) &= \frac{f_Y(y)}{1-F_Y(y)} = \frac{\frac{2e^y}{(1+e^y)^2}}{1 - \frac{e^y-1}{1+e^y}} \\ &= \frac{1+e^y}{2} \times \frac{2e^y}{(1+e^y)^2} = \frac{e^y}{1+e^y} \end{aligned}$$

(5)

$$x = \frac{y-\mu}{\sigma}$$

$$Y = \sigma X + \mu = g(x)$$

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

$$= f_X\left(\frac{y-\mu}{\sigma}\right) \cdot \frac{1}{\sigma}$$

$$= \frac{e^{-\frac{y-\mu}{\sigma}}}{(1+e^{-\frac{y-\mu}{\sigma}})^2} \cdot \frac{1}{\sigma}$$

P23

関数 $g(\cdot)$ を通る

$X \in Y = g(X)$ に変換

Y の分布は X の分布から導出される

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(g(X) \leq y) \\ &= P(X \in \{x \mid g(x) \leq y\}) \end{aligned}$$

X が連続なら

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) \\ &= \frac{d}{dy} P(X \in \{x \mid g(x) \leq y\}) \end{aligned}$$

g が単調増加 $\rightarrow g^{-1}(\cdot)$ が存在

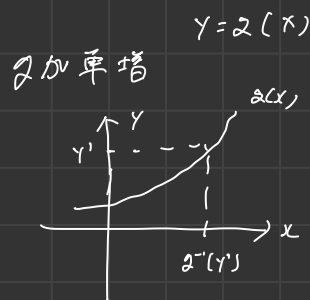
$$\begin{aligned} F_Y(y) &= P(X \in \{x \mid x \leq g^{-1}(y)\}) \\ &= \int_{-\infty}^{g^{-1}(y)} f_X(x) dx \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} \int_{-\infty}^{g^{-1}(y)} f_X(x) dx \\ &= f_X(g^{-1}(y)) \cdot \frac{dg^{-1}(y)}{dy} \end{aligned}$$

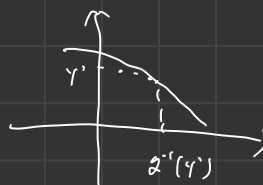
$$g(g^{-1}(y)) = y \quad \frac{1}{dy} = \frac{1}{dx} \frac{dx}{dy}$$

$$g'(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) = 1 \quad \frac{d}{dy} g^{-1}(y) = \frac{1}{g'(g^{-1}(y))}$$

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{1}{g'(g^{-1}(y))}$$



$$\begin{aligned} g(x) &\leq y' \\ x &\leq g^{-1}(y') \end{aligned}$$



$$\begin{aligned} g(x) &\leq y' \\ g^{-1}(y') &\leq x \end{aligned}$$

$$F_Y(y) = \int_{g^{-1}(y)}^{\infty} f_X(x) dx$$

$$= 1 - \int_{-\infty}^{g^{-1}(y)} f_X(x) dx$$

$$f_Y(y) = -f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

$$g(g^{-1}(y)) = y$$

$$g'(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) = 1$$

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{g'(g^{-1}(y))}$$

$$f_Y(y) = -f_X(g^{-1}(y)) \cdot \frac{1}{g'(g^{-1}(y))}$$

$$g(x) = Y$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(g(X) \leq y)$$

$$= P(X \in \{x \mid g(x) \leq y\})$$

$g(x)$ が単増,

$$F_Y(y) = P(X \in \{x \mid x \leq g^{-1}(y)\})$$

$$= \int_{-\infty}^{g^{-1}(y)} f_X(x) dx$$

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

$$g(g^{-1}(y)) = y$$

$$g'(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) = 1$$

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{g'(g^{-1}(y))}$$

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{1}{g'(g^{-1}(y))}$$

$$Y = F_X(X)$$

$$f_Y(y) = f_X(F_X^{-1}(y)) \cdot \frac{1}{f_X(F_X^{-1}(y))}$$

$$= \frac{1}{f_X}$$

問 14

$$N(0,1)$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

(1)

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} dx$$

$$= \int_{-\infty}^{\infty} x \cdot \underbrace{\frac{x}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}}_{\left(-\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}\right)'} dx$$

$$= \left[x \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{x^2}{2}\right\} dx$$

(2) $Y = X^2$

$Y = g(X)$
 $g'(y) = \pm \sqrt{y}$

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y}{2}\right\} \cdot \frac{1}{\sqrt{y}}$$

$$= \frac{1}{\sqrt{2\pi y}} \exp\left\{-\frac{y}{2}\right\}$$

$$E[Y] = \int_0^{\infty} y \cdot \frac{1}{\sqrt{2\pi y}} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \int_0^{\infty} \sqrt{y} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y}{2}\right\}}_{\left(-\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y}{2}\right\}\right)'}$$

$$= \left[\sqrt{y} \cdot \left(-\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y}{2}\right\}\right) \right]_0^{\infty}$$

$$+ \int_0^{\infty} \frac{1}{2\sqrt{y}} \left(\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y}{2}\right\}\right) dy$$

$$= \int_0^{\infty} \frac{1}{2\sqrt{y}} \exp\left\{-\frac{y}{2}\right\} dy = \frac{1}{2}$$

(3)

$Y = |X|$

$F_X(y) - F_X(-y)$

$f_X(y) \cdot \frac{d}{dy} y - f_X(-y) \cdot \frac{d}{dy} (-y)$

$$f_Y(y) = \frac{d}{dy} P(Y \leq y)$$

$$= f_X(y) + f_X(-y)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\} + \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\}$$

$$= \sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\}$$

$$E[Y] = \int_0^{\infty} y \cdot \sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy$$

$$= \left[-\sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\} \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}}$$

$$E[Y^2] = \int_0^{\infty} y \cdot y \sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy$$

$$= \left[y \cdot \left(-\sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\}\right) \right]_0^{\infty}$$

$$+ \int_0^{\infty} \sqrt{\frac{2}{\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy$$

$$= 1$$

$$\text{Var}(X) = 1 - \frac{2}{\pi}$$

問15

$$X \sim N(0, 1)$$

$$M_X(t) = \exp\left\{\frac{t^2}{2}\right\}$$

$$M_X'(0) = \left. t \exp\left\{\frac{t^2}{2}\right\} \right|_{t=0}$$

$$= 0$$

$$M_X''(0) = \left. \exp\left\{\frac{t^2}{2}\right\} + t^2 \exp\left\{\frac{t^2}{2}\right\} \right|_{t=0}$$

$$= \left. (1+t^2) \exp\left\{\frac{t^2}{2}\right\} \right|_{t=0}$$

$$= 1$$

$$M_X^{(3)}(0) = \left. t \cdot \exp\left\{\frac{t^2}{2}\right\} + 2t^3 \exp\left\{\frac{t^2}{2}\right\} + t^5 \exp\left\{\frac{t^2}{2}\right\} \right|_{t=0}$$

$$= \left. (3t + t^5) \exp\left\{\frac{t^2}{2}\right\} \right|_{t=0}$$

$$= 0$$

$$M_X^{(4)}(0) =$$

$$(3 + 3t^2 + 3t^4 + t^6)$$

$$= (t^6 + 6t^2 + 3)$$

$$(4t^3 + 12t + t^5 + 6t^3 + 3t)$$

$$= (t^5 + 10t^3 + 14t)$$

$$E[X^{\frac{p}{2}}] = 2^{\frac{p}{2}} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)}$$

$$= 2^{\frac{p}{2}} \frac{\int_0^\infty y^{\frac{p-1}{2}} e^{-y} dy}{\int_0^\infty y^{-\frac{1}{2}} e^{-y} dy}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

$$E[X^{\frac{p}{2}}] = \int_{-\infty}^{\infty} x^{\frac{p}{2}} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} dx$$

(i) $\frac{p}{2}$ が奇数の場合

$$x^{\frac{p}{2}} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} \text{ は奇関数。}$$

$$\therefore E[X^{\frac{p}{2}}] = \int_{-\infty}^{\infty} x^{\frac{p}{2}} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} dx = 0$$

(ii) $\frac{p}{2}$ が偶数の場合

$Y = X^2$ と変数変換を行うと

$$Y \sim \chi^2_1 \quad Y = X^2$$

$$E[X^{\frac{p}{2}}] = E[Y^{\frac{p}{4}}]$$

$$= \int_0^\infty y^{\frac{p}{4}} \cdot \frac{1}{\Gamma\left(\frac{1}{2}\right)} \left(\frac{1}{2}\right)^{\frac{1}{2}} y^{-\frac{1}{2}} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \frac{1}{\Gamma\left(\frac{1}{2}\right)} \int_0^\infty \left(\frac{1}{2}\right)^{\frac{1}{2}} y^{\left(\frac{p}{4}\right)-\frac{1}{2}} e^{-\frac{y}{2}} dy$$

$$= \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)} \left(\frac{1}{2}\right)^{\frac{p+1}{2}} \int_0^\infty y^{\left(\frac{p+1}{2}\right)-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)} \cdot 2^{\frac{p}{2}}$$

問 16

$$X \sim N(0,1)$$

$$(1) \quad k \in \mathbb{R}^+$$

$$Z = |X|^k$$

$$F_Z(z) = P(Z \leq z)$$

$$= P(|X|^k \leq z)$$

$$= P(|X| \leq z^{\frac{1}{k}})$$

$$= P(-z^{\frac{1}{k}} \leq X \leq z^{\frac{1}{k}})$$

$$= \int_{-z^{\frac{1}{k}}}^{z^{\frac{1}{k}}} f_X(x) dx$$

$$f_Z(z) = f_X(z^{\frac{1}{k}}) \cdot \frac{d}{dz} z^{\frac{1}{k}} \\ - f_X(-z^{\frac{1}{k}}) \cdot \frac{d}{dz} (-z^{\frac{1}{k}})$$

$$= \frac{1}{k} z^{\frac{1}{k}-1} (f_X(z^{\frac{1}{k}}) + f_X(-z^{\frac{1}{k}}))$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

$$f_Z(z) = \frac{2}{k} z^{\frac{1}{k}-1} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^{\frac{2}{k}}}{2}\right\}$$

$$z > 0$$

(2)

$$E[X^{2m}] = 2^m \frac{\Gamma(m+\frac{1}{2})}{\Gamma(\frac{1}{2})} = \frac{2^m}{\sqrt{\pi}} \Gamma(m+\frac{1}{2})$$

$$E[|X|^{2m+1}]$$

$$= E[Z]$$

$$= \int_0^\infty z \cdot \frac{2}{k} \cdot z^{\frac{1}{k}-1} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^{\frac{2}{k}}}{2}\right\} dz$$

$$= \frac{2}{\sqrt{2\pi} k} \int_0^\infty z^{\frac{1}{k}} \exp\left\{-\frac{z^{\frac{2}{k}}}{2}\right\} dz$$

$$z^{\frac{1}{k}} = t \quad \frac{dt}{dz} = \frac{1}{k} z^{\frac{1}{k}-1}$$

$$dt = \frac{1}{k} \cdot \frac{z^{\frac{1}{k}}}{z} dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^\infty t^{2m+1} \exp\left\{-\frac{t^2}{2}\right\} dt$$

$$u = \frac{t^2}{2} \quad \frac{du}{dt} = t$$

$$2u = t^2$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^\infty (2u)^m e^{-u} du$$

$$= \frac{2}{\sqrt{2\pi}} \cdot 2^m \int_0^\infty u^{(m+1)-1} e^{-u} du$$

$$= \frac{2^{m+1}}{\sqrt{2\pi}} \Gamma(m+1) = \frac{2^{m+1}}{\sqrt{2\pi}} m!$$

Ex 17

$$f(x) = c(\alpha) e^{-|x|^\alpha}, \quad -\infty < x < \infty$$

$$Y = |X|^\alpha$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(|X|^\alpha \leq y)$$

$$= P(|X| \leq y^{\frac{1}{\alpha}})$$

$$= P(-y^{\frac{1}{\alpha}} \leq X \leq y^{\frac{1}{\alpha}})$$

$$\begin{aligned} f_Y(y) &= f_X(y^{\frac{1}{\alpha}}) \cdot \frac{1}{\alpha} y^{\frac{1}{\alpha}-1} \\ &\quad + f_X(-y^{\frac{1}{\alpha}}) \cdot \frac{1}{\alpha} y^{\frac{1}{\alpha}-1} \\ &= \frac{2}{\alpha} y^{\frac{1}{\alpha}-1} \cdot c(\alpha) \cdot e^{-y} \end{aligned}$$

$$1 = \int_0^\infty \frac{2}{\alpha} y^{\frac{1}{\alpha}-1} c(\alpha) \cdot e^{-y} dy$$

$$= c(\alpha) \cdot \frac{2}{\alpha} \int_0^\infty y^{\frac{1}{\alpha}-1} \cdot e^{-y} dy$$

$$= c(\alpha) \cdot \frac{2}{\alpha} \cdot \Gamma\left(\frac{1}{\alpha}\right)$$

$$\therefore c(\alpha) = \frac{\alpha}{2\Gamma\left(\frac{1}{\alpha}\right)}$$

$$E[|X|^r] = E[Y^{\frac{r}{\alpha}}]$$

$$= \int_0^\infty y^{\frac{r}{\alpha}} \cdot \frac{2}{\alpha} \cdot \frac{\alpha}{2\Gamma\left(\frac{1}{\alpha}\right)} \cdot y^{\frac{1}{\alpha}-1} \cdot e^{-y} dy$$

$$= \frac{1}{\Gamma\left(\frac{1}{\alpha}\right)} \int_0^\infty y^{\frac{r+1}{\alpha}-1} e^{-y} dy$$

$$= \frac{\Gamma\left(\frac{r+1}{\alpha}\right)}{\Gamma\left(\frac{1}{\alpha}\right)}$$

✓

問 18

$$X \sim \chi_n^2$$

(1)

$$\begin{aligned} E[X^\sigma] &= \int_0^\infty \frac{x^\sigma}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} x^{\frac{n}{2}-1} \exp\left\{-\frac{x}{2}\right\} dx \\ &= \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} \int_0^\infty x^{(\frac{n}{2}+\sigma)-1} \exp\left\{-\frac{x}{2}\right\} dx \end{aligned}$$

$$y = \frac{x}{2} \quad \frac{dy}{dx} = \frac{1}{2}$$

$$= \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} \int_0^\infty 2^{(\frac{n}{2}+\sigma)} \cdot y^{(\frac{n}{2}+\sigma)-1} e^{-y} dy$$

$$= 2^\sigma \frac{\Gamma(\frac{n}{2}+\sigma)}{\Gamma(\frac{n}{2})}$$

(2)

$$E[h(X_n^2)]$$

$$f_{X_n^2}(x) = \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} x^{\frac{n}{2}-1} \exp\left\{-\frac{x}{2}\right\}$$

$$f_{X_{n+2}^2}(x) = \frac{1}{\Gamma(\frac{n}{2}+1)} \left(\frac{1}{2}\right)^{\frac{n}{2}+1} x^{\frac{n}{2}+1-1} \exp\left\{-\frac{x}{2}\right\}$$

$$f_{X_n^2} = \frac{1}{\frac{n}{2} \Gamma(\frac{n}{2}+1)} \left(\frac{1}{2}\right)^{-1} \cdot \left(\frac{1}{2}\right)^{\frac{n}{2}+1} \cdot x^{-1} \cdot x^{\frac{n}{2}+1-1} \exp\left\{-\frac{x}{2}\right\}$$

$$= \frac{n}{x} f_{X_{n+2}^2}(x)$$

$$\begin{aligned} E[h(X_n^2)] &= \int_0^\infty h(x) f_{X_n^2}(x) dx \\ &= \int_0^\infty h(x) \cdot \frac{n}{x} f_{X_{n+2}^2}(x) dx \end{aligned}$$

$$= n \int_0^\infty \frac{h(x)}{x} f_{X_{n+2}^2}(x) dx$$

$$= n E\left[\frac{h(X_{n+2}^2)}{X_{n+2}^2}\right]$$

$$\begin{aligned} f_{X_{n+2}^2}(x) &= \frac{1}{\Gamma(\frac{n}{2}+1)} \left(\frac{1}{2}\right)^{\frac{n}{2}+1} x^{\frac{n}{2}+1-1} \exp\left\{-\frac{x}{2}\right\} \\ &= \frac{(\frac{n}{2}-1)}{\Gamma(\frac{n}{2})} \end{aligned}$$

$$f_{X_n^2}(x) = \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} x^{\frac{n}{2}-1} \exp\left\{-\frac{x}{2}\right\}$$

$$= \frac{1}{\Gamma(\frac{n}{2}-1)} \cdot \frac{1}{(\frac{n}{2}-1)} \cdot \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right)^{\frac{n}{2}-1} \cdot x \cdot x^{\frac{n}{2}-1-1} \exp\left\{-\frac{x}{2}\right\}$$

$$= \frac{x}{n-2} f_{X_{n-2}^2}(x)$$

$$E[h(X_n^2)] = \int_0^\infty h(x) \cdot f_{X_n^2}(x) dx$$

$$= \int_0^\infty h(x) \cdot \frac{x}{n-2} f_{X_{n-2}^2}(x) dx$$

$$= \frac{1}{n-2} \int_0^\infty x \cdot h(x) f_{X_{n-2}^2}(x) dx$$

$$= \frac{1}{n-2} E[X_{n-2}^2 h(X_{n-2}^2)]$$

問19

$$X \sim \text{Bin}(n, p)$$

$$Y \sim \text{Beta}(n-x, x+1)$$

($0 \leq x \leq n-1$ の整数)

$$P(X \leq x) = P(Y \leq 1-p) \text{ を示す.}$$

$$P(X \leq x) = \sum_{k=0}^x \binom{n}{k} p^k \cdot (1-p)^{n-k}$$

$$P(Y \leq 1-p) = \int_0^{1-p} \frac{1}{B(n-x, x+1)} y^{n-x-1} (1-y)^{x+1-1} dy$$

$$P(X \leq x) = nC_0 p^0 (1-p)^n + nC_1 p^1 (1-p)^{n-1} + \dots + nC_x p^x (1-p)^{n-x}$$

$$\begin{aligned} P(Y \leq 1-p) &= \frac{\Gamma(n+1)}{\Gamma(n-x)\Gamma(x+1)} \int_0^{1-p} y^{n-x-1} (1-y)^{x+1-1} dy \\ &= \frac{n!}{(n-x)!x!} \int_0^{1-p} \left(\frac{1}{n-x} y^{n-x} \right) (1-y)^{x+1-1} dy \\ &= \frac{n!}{(n-x)!x!} \int_0^{1-p} (y^{n-x}) (1-y)^{x+1-1} dy \\ &= \frac{n!}{(n-x)!x!} \left\{ \left[y^{n-x} (1-y)^{x+1-1} \right]_0^{1-p} + \int_0^{1-p} y^{n-x} \cdot x (1-y)^{x-1} dy \right\} \\ &= \frac{n!}{(n-x)!x!} \left\{ (1-p)^{n-x} p^x + \int_0^{1-p} y^{n-x} x (1-y)^{x-1} dy \right\} \\ &= nC_x (1-p)^{n-x} \cdot p^x \\ &\quad + \frac{n!}{(n-x)! \cdot (x-1)!} \int_0^{1-p} y^{n-x} (1-y)^{x-1} dy \end{aligned}$$

$$= nC_x (1-p)^{n-x} \cdot p^x + \int_0^{1-p} \frac{1}{B(n-(x-1), (x-1)+1)} y^{n-(x-1)-1} (1-y)^{(x-1)+1-1} dy$$

$$= nC_x (1-p)^{n-x} \cdot p^x + nC_{x-1} (1-p)^{n-(x-1)} \cdot p^{(x-1)} + \dots + nC_0 (1-p)^n \cdot p^0$$

$$= P(X \leq x)$$

$$B(n-(x-1), (x-1)+1)$$

$$= B(n-x+1, x)$$

$$= \frac{\Gamma(n-x+1) \Gamma(x)}{\Gamma(n+1)}$$

$$= \frac{(n-x)! \cdot (x-1)!}{n!}$$

Prob 20

$$X \sim \text{Po}(\lambda)$$

$$\begin{aligned} P(X \leq x-1) &= \sum_{k=0}^{x-1} \frac{\lambda^k}{k!} \cdot e^{-\lambda} \\ &= \frac{\lambda^0}{0!} e^{-\lambda} + \frac{\lambda^1}{1!} e^{-\lambda} \\ &\quad + \dots + \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} \end{aligned}$$

$$Y \sim \text{Ga}(x, 1) \quad np = \lambda$$

$$\begin{aligned} P(Y \geq \lambda) &= \int_{\lambda}^{\infty} \frac{1}{\Gamma(x)} \cdot y^{x-1} \cdot e^{-y} dy \\ &= \frac{1}{(x-1)!} \int_{\lambda}^{\infty} y^{x-1} e^{-y} dy \\ &= \frac{1}{(x-1)!} \int_{\lambda}^{\infty} y^{x-1} \cdot (-e^{-y})' dy \\ &= \frac{1}{(x-1)!} \left\{ \left[y^{x-1} \cdot (-e^{-y}) \right]_{\lambda}^{\infty} + (x-1) \int_{\lambda}^{\infty} y^{x-2} \cdot (-e^{-y}) dy \right\} \\ &= \frac{1}{(x-1)!} \left\{ \frac{\lambda^{x-1}}{e^{\lambda}} + (x-1) \cdot \int_{\lambda}^{\infty} y^{x-2} \cdot e^{-y} dy \right\} \\ &= \frac{1}{(x-1)!} \cdot \frac{\lambda^{x-1}}{e^{\lambda}} + \frac{1}{(x-2)!} \int_{\lambda}^{\infty} y^{x-2} e^{-y} dy \\ &= \frac{1}{(x-1)!} \cdot \frac{\lambda^{x-1}}{e^{\lambda}} + \int_{\lambda}^{\infty} \frac{1}{\Gamma(x-1)} \cdot y^{(x-1)-1} \cdot e^{-y} dy \\ &= \frac{1}{(x-1)!} \cdot \frac{\lambda^{x-1}}{e^{\lambda}} + \frac{1}{(x-2)!} \cdot \frac{\lambda^{x-2}}{e^{\lambda}} \\ &\quad + \dots + \frac{1}{0!} \cdot \frac{\lambda^0}{e^{\lambda}} \\ &= P(X \leq x-1) \end{aligned}$$